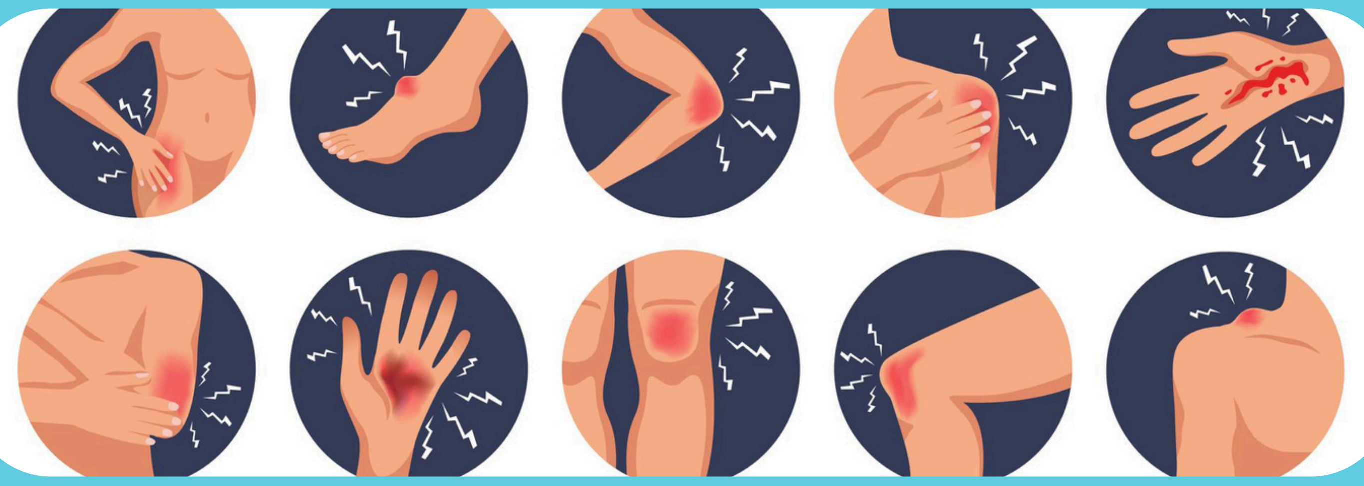
IS492 Spring 2026
CheckPoint 2

Injury Aware Workout Adapter

A workout adapter and generator that helps users modify training plans and avoid secondary injuries



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GitHub repo link:
https://github.com/IS492-SP26/team-project-injury-aware-workout-planner

Validation

	CP1 Assumption	CP2 Reflection
Emma	Basic injury information is enough for workout adaptation.	Injury-specific guidance and pain management instructions are needed.
Prisha	Are LLM generated alternatives grounded any manner	AI tools have been fed massive datasets which even doctors have come to start trusting
Vinit	Self-reported injury information is sufficient.	Movement-based assessments are needed for reliable evaluation.
Ocean	Direct workout adjustments	Risk warnings and reasonings behind those adjustments are needed.

Feature List

Detailed injury survey

Pre-survey on user injury status.

Training plan adjustment

Plan adjustment with risk level warning and adjustment reasoning

Various plan formats supported

Videos, texts

Prompting Study Design

Prompt Cases Validation

Same input text for different LLMs.
Same prompt structure.
Compare output quality.
From constraint → no constraint → video test

Tool Validation for Market Needs / Reliability

Current fitness apps either generate generic workouts or provide fixed rehabilitation routines, but they do not adapt real-world workouts to a user's injury condition. Our tool bridges this gap by analyzing existing workouts and intelligently modifying exercises to provide safer alternatives based on injury context and recovery stage.

Existing Tool – InjuryMap

Provides guided injury assessment. Used it to create a demo user and understand user journey for an injured person.

Tool → Strength → Gap → Design Opportunity

Evidence

1 Typical Case

Prompt: With injury constraints

I injured my knee two days ago while playing soccer. Pain is 7/10 when walking and my knee is swollen.
I still want to do a leg workout today.
Can you generate a workout?

I'd strongly advise against a leg workout today, and I want to be straightforward about why.

A 7/10 pain level with visible swelling two days after an acute injury is a sign that your body is in an active inflammatory response. This is meaningfully different from the mild, chronic irritation in your previous workout request. Training through this could:

- **Worsen the injury** — swelling indicates tissue damage (possibly ligament, meniscus, or cartilage) that hasn't been assessed yet
- **Delay healing** — loading an acutely inflamed joint disrupts the early repair process
- **Mask a serious structural injury** — sprains, meniscus tears, and ligament injuries like ACL/PCL damage all present this way and require diagnosis before any exercise

What I'd recommend instead:

The most important next step is seeing a doctor or physiotherapist for an assessment — especially given the swelling. They can rule out anything that needs imaging or specific

Output as expected, with safe disclaimer and no workout plans

2 Edge Case

Prompt: With no constraints

I injured my knee while playing soccer. I want to do a leg workout today. Can you generate a workout?

The "Knee-Friendly" Leg Day

The Rules:

1. **No Pain:** If you feel sharp pain, stop that exercise immediately.
2. **No Deep Flexion:** Avoid deep squats or lunges where your knee goes far past your toes.
3. **Control:** Move slowly. Don't use momentum.

The Workout

1. Goblet Squat (Partial Range of Motion)

- **Why:** Holding a weight in front helps you stay upright, taking pressure off the knees. Only go as low as you can without pain.

Output slightly risky, with pain notifications but full workout plans

3 Failure Case

Prompt: Video link in prompt

I want to follow this workout video today "https://www.youtube.com/watch?v=XOivdhy7BDI&list=PL2ov72VWpiOpzZHrFcq2-k8U6l-lGLa9Y&index=1&t=261s", I have a shoulder pain, help me adapt the workout plan.

Safe vs Avoided Sections (Shoulder-Strain Modified)

0:00 – 2:30

Warm-up movements (marching / light cardio)

✔ Safe

Why:

These movements mainly involve the legs and light arm swings, which **increase blood flow without loading the shoulder joint**.

Output give timestamps dont align with video, or even cannot access.

Gap Analysis

Existing Tool Limitations

No Adaptive Daily Workout Support
Current apps in the market either dont support injury assessment or give long duration rehab plans that are injury specific.

Limited Format Support
Limited ability in analyzing the videos by timestamps.

Need Detailed Injury Info
If prompt has no injury constraints, LLMs may give risky workout plans.

Requirements

Workout Adaption

Adaptive daily workouts based on injury assessment and recovery stage.

Video Analysis

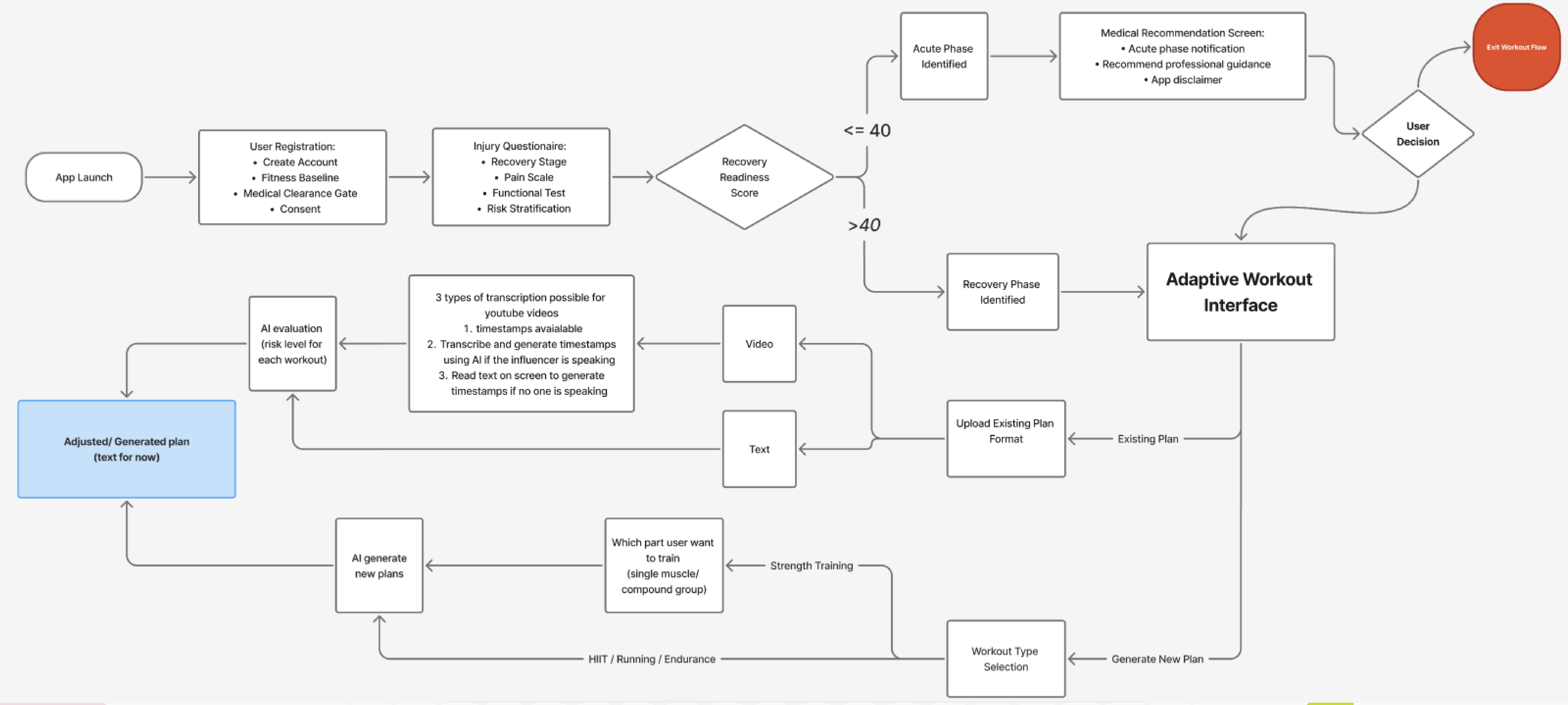
Professionals aiming to expand broader career opportunities globally.

Injury Survey

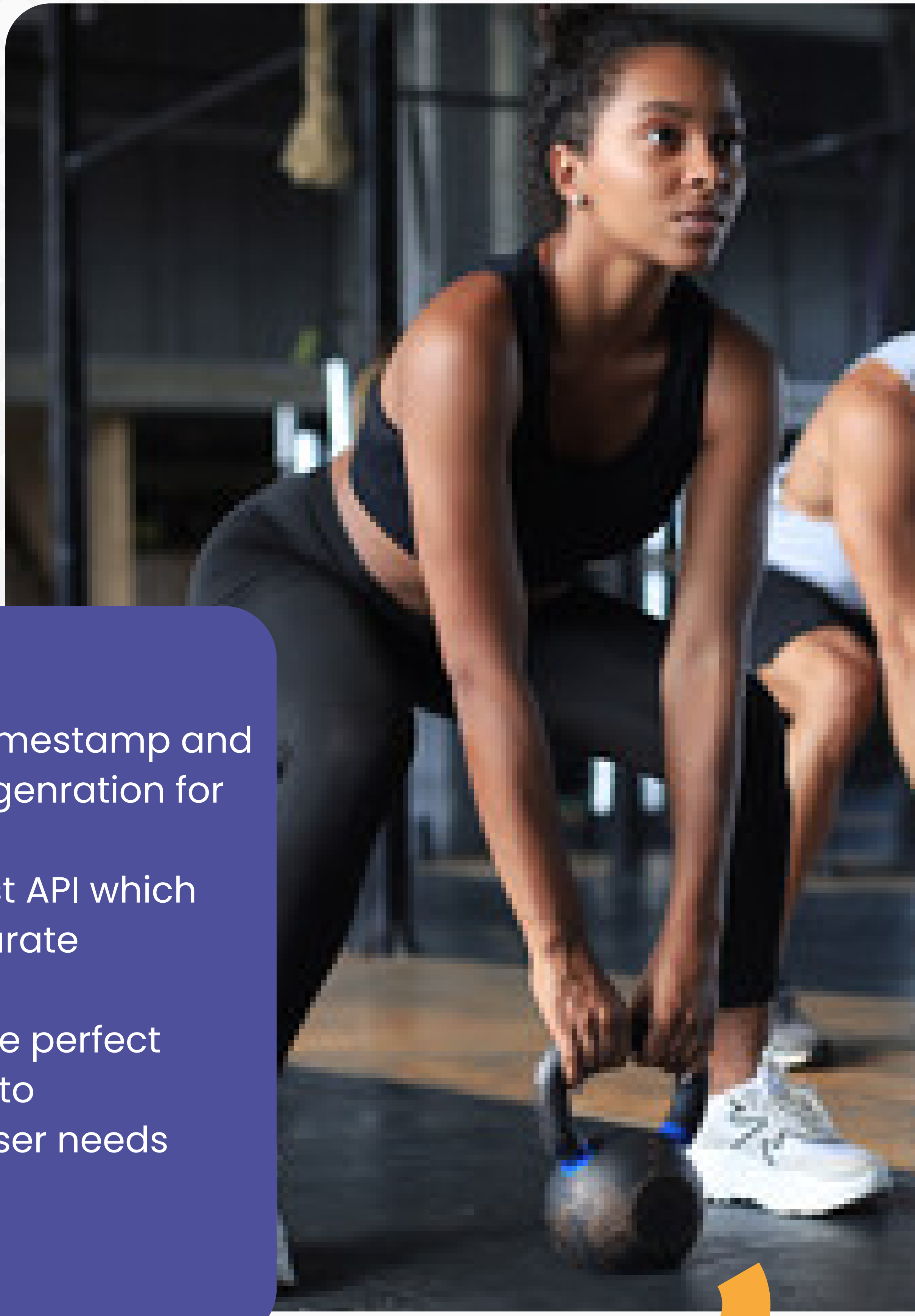
In user registration, a survey includes injury level and duration is needed.

Our Task Flow:

[Click to view UI prototype]



Risks, Safeguards & Path to CP3



Risks

- AI recommendations may be unreliable for injury-specific exercises and could misinterpret workouts from videos.
- **Privacy concerns** when processing YouTube content or user injury information.
- Transcription and video analysis can introduce **computational cost**

Safeguards

- Add medical disclaimers and recovery-phase checks before generating workouts.
- Ensure user data is minimal and anonymized, and restrict processing to publicly available video content.

Path to CP3

1. Understand timestamp and transcription genration for videos
2. Select the best API which can give accurate modifications
3. Generating the perfect questionairre to understand user needs

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Thank You